Good.
Two weeks is tight — but possible if you are disciplined (2–3 hrs/day).

I will give you:

1. **What to study**
2. **Exact order**
3. **Free high-quality sources**
4. **What level you must reach**

No overload. Only what matters for this Micron role.

🔥 PART 1 — C Language Preparation (Critical)

For this JD, C is not DSA-heavy.
It is **memory + pointer + system-level C**.

🎯 You Must Master These:

1️⃣ Pointers (Non-Negotiable)

- Pointer basics
- Pointer arithmetic
- Pointer to pointer
- Passing pointer to function
- Function pointers
- Array vs pointer difference

📚 Source:

- YouTube: **"Neso Academy – Pointers in C"**
- Practice: solve pointer problems on HackerRank (C track)
- Book reference (free PDF searchable): *"C Programming: A Modern Approach – K.N. King"* (pointer chapters)

2️⃣ Memory Management

- malloc
- calloc
- realloc
- free
- memory leak

- dangling pointer
- NULL handling

Practice:

- Implement dynamic array
- Implement linked list
- Free memory properly

📚 Source:

- GeeksforGeeks (malloc & free explained well)
- Write and run on local GCC

3️⃣ Structures & Memory Layout

- Structure padding
- sizeof behavior
- Alignment
- Bit fields

Why?
Because firmware uses register structures.

📚 Source:

- YouTube: “Structure Padding in C – Jenny’s Lectures”
- Write small code and check sizeof output

4️⃣ Bit Manipulation

- Set bit
- Clear bit
- Toggle bit
- Check bit
- Bit masks

Essential for firmware validation.

📚 Source:

- GeeksforGeeks Bitwise Operators in C
- Practice writing macros

5️⃣ Debugging C with gdb

- Compile with -g
- Breakpoint
- Print variable
- Backtrace
- Segmentation fault debugging

📚 Source:

- YouTube: “GDB Tutorial for Beginners”
- Or: Tutorialspoint GDB basics

🔥 PART 2 — NVMe + SSD (Concept Level Only)

You don’t need spec-level mastery.

🎯 **Learn Only This:**

1️⃣ What is SSD internally?

- NAND flash
- Pages
- Blocks
- Wear leveling
- Garbage collection
- FTL mapping

📚 Source:

- YouTube: “SSD Architecture Explained”
- Search: “FTL explained simple”

2️⃣ NVMe Architecture

Learn:

- Submission Queue
- Completion Queue
- Doorbell registers

- Admin vs IO commands
- NVMe read/write flow

📚 Source:

- YouTube: “NVMe basics explained”
- Western Digital blog on NVMe basics
- Simplified NVMe intro articles

Do NOT read full spec.

🔥 **PART 3 — Linux Debugging (Important)**

Must Learn:

- dmesg
- journalctl
- ps / top
- kill
- /proc
- tail -f
- strace (basic idea)
- segmentation fault meaning

📚 Source:

- “Linux Command Line – William Shotts” (free PDF)
- YouTube: “Linux troubleshooting basics”

Practice daily in Ubuntu VM.

🔥 **PART 4 — Python Automation (Minimal but Required)**

You only need:

- Read file
- Regex
- Count patterns
- Write summary
- Exception handling

Mini Project:

- Parse log file
- Extract “ERROR”
- Count
- Print PASS if <5 errors

📚 Source:

- “Corey Schafer Python File Handling” (YouTube)
- Regex tutorial Python

🗓️ **14-Day Focus Plan**

Days 1–5

C Pointers + Memory + Structures

Days 6–7

Bit manipulation + gdb practice

Days 8–9

Linux debugging tools

Days 10–11

SSD + FTL concepts

Days 12–13

NVMe architecture basics

Day 14

Write small automation script + revise everything

🎯 **After 2 Weeks You Should Be Able To:**

- Explain stack vs heap clearly
- Debug segmentation fault
- Explain FTL confidently
- Draw NVMe queue diagram
- Write basic log parser
- Talk about firmware validation strategy

Then your rating improves to:

C → 6.5–7
Linux → 6.5
NVMe → 5.5–6
Automation → 6
Overall → 7/10 for Associate

⚠ **Important**

Two weeks means:

- No Instagram
- No random YouTube
- No switching topics
- Daily discipline